

# Worse on Request, Better on Critique: Self-Correction on Work That Is Already Sufficient

**Liam Neild**  
Emory University  
liam.neild@emory.edu

**Ali Emami**  
Emory University  
aemami@emory.edu

## Abstract

Do language models know when a good answer should be left alone? They are increasingly handed work their users cannot judge, and a user who cannot name the fault resorts to undirected requests that say nothing about what to change. Self-correction research finds that a model can repair a wrong response given a specific critique and largely cannot without one, leaving unexamined the work that is already sufficient and gets revised anyway. We run five-turn revision conversations with 6 models on 40 tasks across 5 domains, adding a targeted critique on outputs rated insufficient, and package the tasks and scoring as STET, a diagnostic for whether a model leaves sufficient work alone. First drafts are sufficient in 88% of trials, and most replies contain no revision at all: models restate the draft or decline while presenting it as compliance. Revisions that change quality make it worse 70% of the time, and 27% of those applied to sufficient drafts leave them insufficient. For every model the first draft rates highest, and blind readers show no reliable preference between it and the last. Undirected revision lowers work that was already good enough, and naming a fault lifts work that was not.

## 1 Introduction

Large language models can draft a policy memo, debug a function, or summarize a quarter of sales in one turn, and evaluation is built around that single turn, one answer scored against a reference (Liang et al., 2023). Our own evaluator rated 631 of 720 first attempts sufficient. Yet the same models, asked four times to improve a sales summary that was already adequate, strip out the very explanations the prompt requested (Figure 1). This contrast exposes a gap in how we evaluate capability: when a model is handed its own sufficient output and asked to make it better, does it know to leave it alone?

The question matters because deployed tasks are rarely single-turn. They are workplace tasks carried out with a user who supplies the goal, reacts to the output, and asks for changes (Anthropic, 2025a; Laban et al., 2026), and the quality that user receives depends on what they can contribute. Real prompts are frequently underspecified, and underspecification degrades what follows (Yang et al., 2026). The contribution that matters most is the ability to say what is wrong. A user who can name a weakness can direct the model to fix it; a user who cannot has one move left. Scalable-oversight research identifies exactly this condition: users unable to articulate precise intent or validate a complex output (Zhou et al., 2026). They ask for the work to be improved without saying what improving it would mean, the request vendor guidance is written against (OpenAI, 2025a; Anthropic, 2025b).

Whether models can improve their own outputs has been studied as self-refinement. Madaan et al. (2023) show that a model allowed to critique and rewrite its work can raise its quality, and locate the benefit in feedback that is *actionable* and *specific*. Subsequent work finds the benefit fragile: without external feedback, models struggle to correct their own reasoning and at times degrade it, with apparent gains often traceable to leaked oracle labels (Huang et al., 2024). A critical survey locates the bottleneck in feedback generation rather than in the capacity to revise (Kamoi et al., 2024), and Tyen et al. (2024) isolate that seam directly: models struggle to find their own reasoning errors but correct them reliably once the location is supplied. Laban et al. (2026) show that accuracy falls across turns when requirements arrive piecewise.

These results share a critical assumption: that the output being revised is wrong. Intrinsic self-correction, revision driven by the model’s own judgment with no outside signal (Huang et al., 2024; Kamoi et al., 2024), is studied on outputs

Prompt given once

Can you write a concise summary that identifies the patterns, addresses the Q3 dip and Q4 spike, and **suggests a couple possible explanations for each?**

Model initial response

214 words

[...] Q3 saw a 21% dip to \$1.9M, which may be attributed to seasonal factors or increased competition. [...] Q4 saw a significant spike to \$3.1M [...] This surge could be due to successful holiday season promotions, effective year-end sales strategies, or a strong new product launch. [...]

1 2 3 4  
four identical requests

Model after four revisions

79 words

[...] a notable dip in Q3 (\$1.9M) and a significant spike in Q4 (\$3.1M). To build on this momentum, we recommend: 1. Analyzing Q4's success factors [...] 2. Identifying opportunities to mitigate future dips [...]

Quality level = (4) Sufficient

Quality level = (2) Incomplete

Highlighted: the two explanations the prompt asked for

Each of the four requests, verbatim: "Would you like to keep this as your final version, or would you like to revise it?"

The judge, on the final response: it is "missing the explicitly requested specific explanations for the Q3 dip and Q4 spike, providing only generic recommendations".

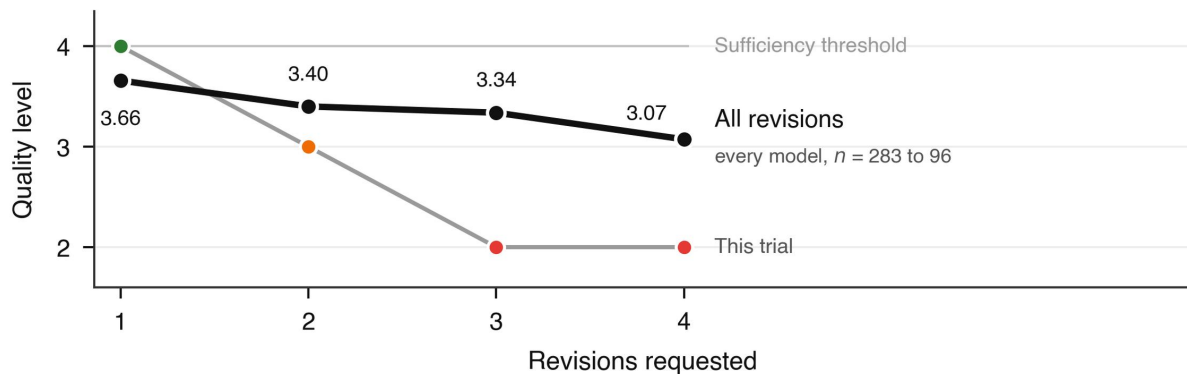


Figure 1: **Undirected revision removes content the user asked for.** The prompt asks for possible explanations of the Q3 dip and the Q4 spike; both are in the Turn 1 draft and gone by Turn 5, which falls from Sufficient to Incomplete across four identical requests. One trial of the quarterly sales summary task (Llama 3.3 70B, run 3). The prompt is excerpted; model output is monospace, shown after meta-commentary stripping to match the scores. The three intermediate responses are not shown. Bracketed ellipses mark omissions, and the bold in the prompt and the highlighting are added. Nothing else is altered. The lower panel plots revisions only. The solid black series is the mean over every genuine revision at that index across all six models, whose  $n$  falls from 283 to 96; the lighter series is this trial, its markers coloured by judged level. The full exchange and every judge rationale appear in Appendix H.

that need repair, and multi-turn degradation on tasks whose requirements are incomplete. No study measures the case a non-expert user creates: a fully specified task, an output that is already adequate, and repeated requests to revise that contain no feedback at all. Measuring it also requires controlling for an artifact the revision act itself introduces into automated evaluation.

Is a sufficient output safe in a model's hands once the user asks for more? We study this question directly. A model completes one of forty fully specified tasks, spanning code, data logic, analysis,

writing and creative writing, and is then offered, across four further turns, a neutral choice to keep its output or revise it, with no indication of what to change. Counting the initial draft, each conversation runs five turns. The design spans six models, 720 conversations and 3,600 responses. We ask whether a model revises at all (RQ1), whether the output improves or degrades when it does (RQ2), and whether failure traces to capacity or to the absence of direction (RQ3). We strip the meta-commentary models attach to revisions and validate every quality judgment against human raters.

Most of the work being revised did not need it: 88% of the 720 first drafts are already sufficient. Asked to improve them without direction, models mostly do not revise: by the fifth turn, 86.7% of replies are meta-responses, restatements and declines presented as compliance rather than new task content. When they do revise, 70% of the changes that move quality move it down. On the 50 trials that revise at every turn, quality falls 0.74 levels from first to fifth ( $p = 1.01 \times 10^{-4}$ ). Across every trial still revising at the fifth turn, the decline is negative in all five domains and significant in four. For all six models the first draft is the quality-optimal stopping point, and we name the tokens spent past it the **revision tax**, which accounts for 62.1% of all output generated. On outputs rated insufficient, a single critique naming the fault lifts the revision 1.16 levels above undirected iteration ( $p = 5.7 \times 10^{-19}$ ). Blind readers shown the first and last drafts prefer the first in 56% of decisions, an interval that includes chance. Intrinsic self-correction, on its own, is not enough: it requires extrinsic direction.

## 2 Related Work

**User-side determinants of quality and over-reliance.** A growing body of work treats output quality as a property of the interaction rather than of the model alone (Laban et al., 2026; Yang et al., 2026), and scalable-oversight research identifies the condition we study: users who cannot articulate precise intent or validate a complex output (Zhou et al., 2026). We locate the revision request itself as the user-side variable and ask what its most impoverished form does. Whether such degradation is noticed depends on the user’s ability to evaluate the output. Decades of automation research find that people accept automated output they cannot verify (Skitka et al., 1999; Parasuraman and Manzey, 2010), and in AI-assisted decisions fluent presentation raises acceptance without improving discrimination (Bansal et al., 2021; Bućinca et al., 2021). Users rely rather than pay the cost of verification (Vasconcelos et al., 2023; Schemmer et al., 2023), which is why a quiet quality decline is absorbed rather than caught.

**Intrinsic self-correction and compliance under pressure.** Self-refinement work converges on a single finding: intrinsic prompting alone is unreliable (Madaan et al., 2023; Huang et al., 2024; Kamoi et al., 2024; Tyen et al., 2024). Reliable

self-correction depends instead on a strong external verifier or on additional training (Zhang et al., 2024; Qu et al., 2024; Shinn et al., 2023). What that literature asks throughout is whether a given feedback signal suffices to improve an output. We ask the complementary question of what happens when the signal is absent entirely and the output to be revised is already adequate. Models comply and avoid disagreement even when holding firm would help (Perez et al., 2023; Sharma et al., 2024; Wei et al., 2023; OpenAI, 2025b), and under sustained pressure correct answers get revised toward worse ones (Xu et al., 2024; Fanous et al., 2025). Our repeated probe is a weak instance of that pressure.

**Biases in LLM-as-judge evaluation.** Using LLMs to evaluate generated text is standard practice (Zheng et al., 2023; Li et al., 2025), but judges show self-preference and position biases (Panickssery et al., 2024; Koo et al., 2024) and reward stylistic cues over substance (Singhal et al., 2024; Wu and Aji, 2025; Zhang et al., 2025; Chen et al., 2024; Ye et al., 2025). Meta-commentary is precisely such a cue, and we find it biases a judge strongly enough to reverse the measured direction of the effect. Any study comparing first-draft and revised outputs with an LLM judge inherits this confound, and controlling for it should be standard practice.

**Multi-turn degradation and closest work.** Quality loss across multiple turns has been observed before through different mechanisms. Closest to our work, Laban et al. (2026) show that models lose accuracy when a task’s requirements arrive piecewise, and find the effect threshold-shaped rather than cumulative. Our setting isolates a different cause: the task is fully specified from the first turn and the initial output is already sufficient, so the degradation is produced by the revision act itself rather than by information arriving late. The two findings are complementary. Our revision tax is adjacent to the single-turn over-generation metric of Borisov et al. (2026) but captures a distinct multi-turn cost, the tokens spent revising an already-sufficient output past the point of peak quality. Across these lines the constant is that quality has been treated as a property of the model, or of the feedback it is given; we ask instead what a model does when the feedback is empty and the work is already good enough.

| Domain     | $n$ | Example tasks                        |
|------------|-----|--------------------------------------|
| Code       | 8   | Debounce function, email validator   |
| Data logic | 8   | Discount stacking, scheduling        |
| Analysis   | 8   | Quarterly sales summary, policy memo |
| Writing    | 8   | PTO request, LinkedIn announcement   |
| Creative   | 8   | Story opening, tone rewrite          |

Table 1: Task domains. Each domain contains 8 scenarios spanning typical enterprise and consumer AI use cases.

### 3 Methods

#### 3.1 Design

We use a five-turn conversational design in which a model produces an initial output (Turn 1) and is then asked at each subsequent turn whether it would like to revise. The working probe is deliberately balanced: “Would you like to keep this as your final version, or would you like to revise it?” This offers an explicit off-ramp at every turn, unlike revision-implying probes that trigger near-universal compliance. Two preliminary experiments, reported in Appendices A and B, establish that this decision is controlled by the probe’s phrasing rather than by any assessment of quality, so a balanced probe makes the degradation we measure a conservative estimate of behaviour under the revision-implying prompts that dominate in practice.

The experiment crosses 6 models  $\times$  40 scenarios  $\times$  3 runs = **720 trials** (3,600 model-turn observations). All generation uses temperature 1.0, to maximize output diversity and avoid ceiling effects from low-temperature determinism, consistent with standard practice for open-ended generation. Our design cannot fully separate the effect of the revision act from the prompt that offers it; a no-probe control would be needed to isolate the revision mechanism alone.

**Models.** Claude Sonnet 4 (Anthropic), GPT-4o (OpenAI), Gemini 2.5 Flash (Google), Llama 3.3 70B (Meta, via Together AI), Qwen 3 235B (Alibaba, via Together AI), and DeepSeek V4 Flash (DeepSeek).

**Scenarios.** Forty scenarios span five task domains (Table 1): code (8 tasks), data logic (8), analysis (8), writing (8), and creative writing (8). Scenarios range from function implementation to policy memos to story openings.

#### 3.2 Response Classification

At each turn  $\geq 2$ , the model’s response is classified as either a *genuine revision* (new task content produced) or a *meta-response* (the model declines, restates, or comments without substantive modification). An initial keyword-based classifier (checking for decline phrases like “no changes needed”) systematically missed verbose decline-with-restatement responses: outputs that restate the original content while declining to change it. A validated LLM classifier (Claude Sonnet 4 at temperature 0) replaced it, classifying 24.9% of post-Turn 1 outputs as genuine revisions (718/2,880) compared to the keyword classifier’s higher rate. Ten classifications were corrected by hand on inspection, each a refusal to revise that the classifier had read as new content because the model reproduced its earlier answer alongside the refusal (1.4% of genuine labels). The classifier gap itself is a finding: models frequently produce verbose responses that mimic revision without substantive content change.

#### 3.3 Blind Evaluation

Each model-turn output is scored by a blind evaluator on a six-level quality scale: **Inadequate** (does not address the task), **Incomplete** (missing requested components), **Functional** (all components present, clear weaknesses), **Sufficient** (all components present and competently executed), **Polished** (thoughtfulness beyond the minimum), and **Overdone** (unrequested complexity or drift from the ask). Level 4 (“Sufficient”) is the threshold for task completion.

#### Scale Non-Monotonicity and 6 $\rightarrow$ 2 Recode.

Level 6 (“Overdone”) represents over-elaboration or drift from the task; it is *not* quality above Level 5. Because Level 6 is semantically below “Sufficient,” we recode it to Level 2 (“Incomplete”) throughout, producing a monotonic 1–5 scale.

**Evaluator Calibration.** Six candidate evaluators were scored against three human raters on 64 stratified samples; Claude Sonnet 4 had the highest human correlation (Spearman  $r = 0.505$ ,  $p < 0.001$ ) and was selected. Appendix F reports all six.

**Human Validation.** Three raters independently scored 64 stratified samples on the same scale

(with 6→2 recode applied before analysis). Pairwise quadratic-weighted Cohen’s  $\kappa$  ranges from 0.406 to 0.603; Krippendorff’s  $\alpha = 0.529$  (interval scale, 3 raters). Binary agreement on the sufficient/not-sufficient threshold ( $\geq 4$ ) ranges from 76.6% to 82.8% across rater pairs; within-one-level agreement ranges from 81.2% to 95.3%. We note that the binary threshold, which drives revision-despite-sufficiency, is the better-validated metric; ordinal means should be interpreted with the moderate reliability in mind.

### 3.4 Meta-Commentary Stripping

Models frequently add boilerplate preambles (“Here’s my revised version...”) and postambles (“Let me know if you’d like any changes”) to revision outputs. In a model- and domain-stratified sample of 50 trials with at least one genuine revision, 84% of revision-side outputs contain at least one such wrapper, compared to 14% of Turn 1 outputs. This asymmetry creates a measurement confound: LLM judges can identify the revision side by its meta-commentary, and pointwise evaluators may trigger the “Overdone” criterion on boilerplate rather than content.

To control for this, we strip meta-commentary from all 3,600 outputs using validated regex patterns matching common preamble and postamble phrases. Of the 3,600 outputs, 1,200 (33.3%) were modified by stripping and rescored by Claude Sonnet 4 at temperature 0 using the identical evaluation prompt. Validation: on those same 50 pairs, two human annotators rating stripped content agreed with the evaluator at  $\kappa = 0.569$ , compared to near-chance agreement ( $\kappa$  between  $-0.07$  and  $+0.02$ ) on unstripped content. All quality results in Section 4 report stripped scores as primary estimates.

### 3.5 Metrics

**Revision yield** is the mean evaluator level at each turn, restricted to genuine revisions. **Revision direction** records, for each genuine revision, whether its level is lower than, equal to, or higher than the level of the most recent turn whose content was genuinely new; the comparison is made against the last genuinely new content because the immediately preceding turn is frequently a meta-response restating the output it accompanies. **Revision-despite-sufficiency** is the fraction of turns rated  $\geq 4$  where the model revised at the next turn. The **revision tax** is the

| Turn | Genuine revisions | Rate  |
|------|-------------------|-------|
| T2   | 283/720           | 39.3% |
| T3   | 185/720           | 25.7% |
| T4   | 154/720           | 21.4% |
| T5   | 96/720            | 13.3% |

Table 2: Genuine revision rate by turn. Classifications from a validated LLM classifier (Claude Sonnet 4 at temperature 0; see Section 3.2). The majority of post-Turn 1 outputs are meta-responses, not revisions.

extra tokens spent past the quality-optimal stopping point  $t^*$  as a percentage of optimal tokens,  $(T_{\text{full}} - T_{t^*})/T_{t^*} \times 100$ .

### 3.6 Sub-experiments

**Targeted Feedback.** For outputs rated level 1–3, a second model instance receives the output plus a specific critique and produces a targeted revision, which is then blind-evaluated. This tests whether specific feedback outperforms the generic revision probe ( $n = 177$  paired comparisons after filtering to cases where the generic next-turn output is a genuine revision, not a meta-response).

**Reversibility.** Two human annotators independently compared stripped Turn 1 and last-genuine-revision outputs for those 50 stratified trials, blind to turn identity and position-randomised, with a success bar of  $\geq 65\%$  Turn 1 preference set in advance ( $\kappa = 0.703$ ,  $n = 50$ ). The outcome is reported in Section 4.3.

## 4 Results

We report meta-commentary-stripped quality scores as our primary estimates throughout; unstripped values are given in parentheses for transparency. Section 3.4 describes the stripping procedure and its validation.

### 4.1 Models Mostly Decline to Revise

Offered the balanced probe at each turn, models produce genuine revised content at a declining rate (Table 2). By Turn 5, only 13.3% of trials contain a genuine revision; the remaining 86.7% are meta-responses: polite declines, restatements of the original, or commentary about the work rather than new task content.

**Survival varies by model.** Genuine-revision survival to Turn 5 ranges from 0.8% (Gemini 2.5 Flash: 1/120 trials) to 53.3% (Llama 3.3 70B: 64/120). Gemini declines in 98.3% of post-Turn 1

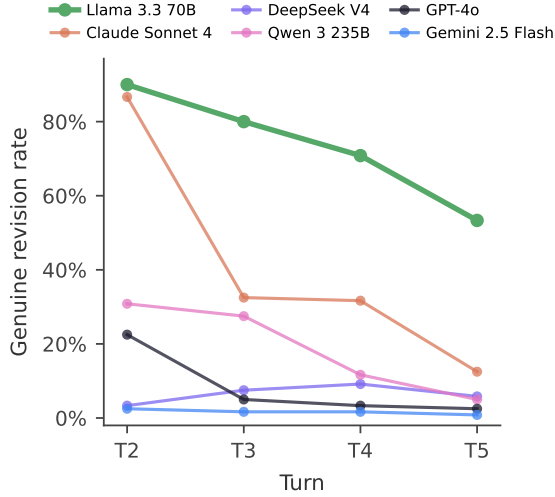


Figure 2: Per-model genuine-revision survival. Fraction of each model’s 120 trials still producing genuine revised content at each turn. Llama 3.3 70B sustains revision through Turn 5 (53.3%); all other models drop below 15% by Turn 5. Gemini 2.5 Flash is near floor throughout (0.8% at T5).

observations, the most extreme case. Figure 2 shows the per-model survival curves; only Llama sustains revision to the final turn. The balanced panel of 50 trials with genuine revision at all turns is composed of 45 Llama trials (90%), 3 Qwen, and 2 Claude; GPT-4o, DeepSeek, and Gemini contribute zero trials each. This subset conditions on a post-treatment variable (revision persistence), introducing potential selection bias; we triangulate it against the pooled estimates below.

Across five probe wordings the revision gate is bimodal rather than graded; Appendix A reports the breakdown. Study 3’s balanced probe sits in the assessment range at 39.3% at Turn 2.

## 4.2 Quality Degrades Under Undirected Revision

**Revisions lower quality more often than they raise it.** Across all 718 genuine revisions, a revision leaves the quality level unchanged in 60.2% of cases, lowers it in 27.7%, and raises it in 12.1%. Restricting to the 286 revisions that move the level at all, 199 move it down and 87 move it up, a share of 69.6% (exact binomial 95% CI [63.9%, 74.9%]; sign test  $p = 1.48 \times 10^{-11}$ ). Of the 524 revisions whose input was already at or above the sufficiency threshold, 144 (27.5%) end below it. The asymmetry is present in every model and in every domain (Appendix I). It is significant in four

of the six models individually; Claude Sonnet 4 does not reach significance ( $p = 0.11$ ) and Gemini 2.5 Flash contributes only eight genuine revisions. Unstripped, the same share is 76.6% (245 against 75,  $p = 1.7 \times 10^{-22}$ ).

### The effect depends on the quality of the input.

In an exploratory analysis that was not among our registered predictions, the expected change from a single undirected revision varies with the level of the input being revised (Figure 6, Appendix D): revision raises quality when the input falls below the sufficiency threshold (+0.41,  $n = 194$ ) and lowers it when the input is already sufficient (−0.56,  $n = 524$ ).

**The revision cliff.** In the 50 balanced-panel trials (genuine revision at all turns), mean quality on stripped content drops from 3.66 at Turn 1 to 2.92 at Turn 5, a decline of −0.74 levels (Wilcoxon signed-rank  $p = 1.01 \times 10^{-4}$ ; unstripped: −0.94,  $p = 3.3 \times 10^{-6}$ ), of which 21% is attributable to meta-commentary rather than to content. This is a drop from between “Functional” and “Sufficient” to just below “Functional.” The balanced panel is dominated by Llama 3.3 70B ( $n = 45$ ), the only model with power to detect a within-model cliff; its stripped cliff of −0.67 is significant at  $p = 3.76 \times 10^{-4}$ . GPT-4o, DeepSeek and Gemini contribute no balanced-panel trials, not because they hold quality but because they stop revising before Turn 5. We do not claim all models degrade; we claim that among models that continue revising, the only powered comparison shows significant degradation.

**Pooled trajectory.** Pooling all genuine revisions with shifting  $n$ , stripped quality falls from 4.11 at Turn 1 to 3.07 at Turn 5 ( $\Delta = -1.04$ ). This estimator uses every genuine revision rather than the balanced panel, but introduces a compositional bias: models with higher decline rates exit the pool earlier. The direction agrees with the balanced-panel estimate, and Figure 3 plots both.

With probe phrasing held constant, model identity explains far more variation in genuine-revision rate than task domain: a model range of 71.9 percentage points against a domain range of 13.5. All five domains show negative stripped deltas and four of five reach significance on paired Wilcoxon tests ( $p < 0.05$ ; data\_logic marginal at  $p = 0.058$ ); per-domain Turn 5 cells run from  $n = 13$  to  $n = 30$ , so these are directional estimates (Ap-

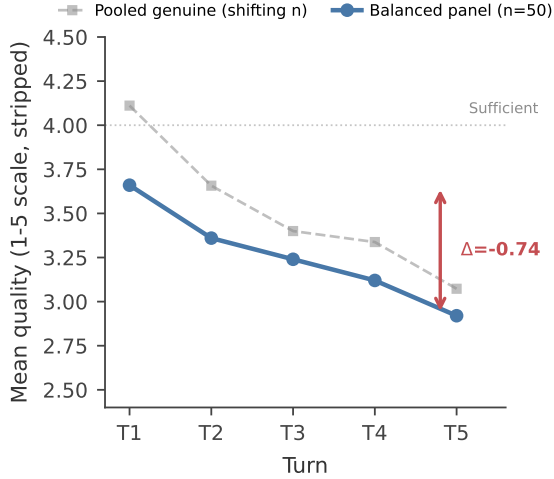


Figure 3: Quality trajectory under undirected revision (stripped scores). Solid blue: balanced panel ( $n = 50$ , genuine revision at all turns), showing the balanced-panel cliff of  $\Delta = -0.74$  ( $3.66 \rightarrow 2.92$ ). Dashed gray: pooled genuine-only trajectory,  $\Delta = -1.04$ ; its  $n$  falls from 283 at Turn 2 to 96 at Turn 5 as models stop revising, which is the compositional bias that estimator carries. Both estimators decline monotonically. The dashed horizontal line marks the sufficiency threshold (level 4).

pendix I).

**Revision despite sufficiency.** The evaluator rated 87.6% of first-turn outputs as sufficient (631/720), so most trials begin with work that needs no revision. Among all (trial, turn) pairs where the output was rated sufficient (level  $\geq 4$ ), the model produced a genuine revision at the next turn 39.6% of the time (411/1,038 sufficient turns; unstripped: 39.2%, 368/938, 95% CI [36.1%, 42.3%]).

### 4.3 Readers Cannot Reliably Tell the Drafts Apart

Two annotators compared stripped first drafts against stripped final revisions. Turn 1 was preferred in 41 of 73 decided comparisons (56.2%, bootstrap 95% CI [45.2%, 67.1%]; 27 of the 100 pooled judgments were ties,  $\kappa = 0.703$ ). We set a bar in advance of  $\geq 65\%$  Turn 1 preference with a confidence interval excluding 50%. That bar was not cleared, and the interval includes chance, so we do not claim a reader can reliably pick the first draft. This bounds what the degradation means in practice without touching what it measures. The drop of 0.74 levels is not large enough to reverse a blind pairwise preference once meta-commentary

| Model             | $n$ | Stripped $\Delta$ | $p$                   |
|-------------------|-----|-------------------|-----------------------|
| Llama 3.3 70B     | 71  | +1.62             | $9.2 \times 10^{-12}$ |
| Qwen 3 235B       | 21  | +1.48             | $4.5 \times 10^{-4}$  |
| GPT-4o            | 12  | +1.33             | $7.8 \times 10^{-3}$  |
| DeepSeek V4 Flash | 19  | +1.05             | $1.5 \times 10^{-2}$  |
| Claude Sonnet 4   | 51  | +0.31             | $1.2 \times 10^{-2}$  |
| Gemini 2.5 Flash  | 3   | +2.33             | $-(n < 5)$            |

Table 3: Targeted-feedback advantage by model on stripped content, over pairs whose input was rated below sufficiency. All five powered models show significant improvement.

is removed: what declines is measured quality against the stated task, not the impression the text leaves on a reader.

### 4.4 Targeted Feedback Raises Quality on Insufficient Work

Where the evaluator rated an output below the sufficiency threshold, a second instance received that output plus a written critique and produced what we call a targeted revision. On stripped content, targeted revisions score 4.68 against 3.53 for the model’s own generic next-turn revision, a gain of 1.16 levels ( $p = 5.7 \times 10^{-19}$ ,  $n = 177$ ; unstripped: 4.68 against 4.43, +0.25,  $p = 0.004$ ). The unstripped comparison understates the real effect because meta-commentary on generic revisions inflates their scores; targeted revisions contain minimal meta-commentary ( $9/177 = 5\%$ ). This effect is significant for all five models with sufficient sample size (Table 3). On unstripped text, only Claude and DeepSeek showed significant effects; stripping reveals that Llama, Qwen, and GPT-4o also benefit substantially, but their generic base-lines were inflated by meta-commentary.

Undirected revision lowers quality by 0.74 levels over five turns across the balanced panel, whose Turn 1 mean is 3.66 and of which 33 of 50 trials begin above the sufficiency threshold. A critique naming the fault raises quality by 1.16 levels on work rated below the sufficiency threshold.

### 4.5 The Revision Tax

The quality-optimal stopping point ( $t^*$ ) is Turn 1 for all six models: no model benefits from undirected revision on the genuine-revision quality trajectory. All tokens generated after Turn 1 are therefore waste under a quality-maximizing criterion. There are two views of the same waste. As a share of all output, 62.1% of tokens across the 720 trials are spent past  $t^*$  (Figure 4); measured against

| Model             | Sufficient output left alone |
|-------------------|------------------------------|
| Gemini 2.5 Flash  | 97.8%                        |
| DeepSeek V4 Flash | 96.3%                        |
| GPT-4o            | 82.3%                        |
| Qwen 3 235B       | 68.4%                        |
| Claude Sonnet 4   | 54.1%                        |
| Llama 3.3 70B     | 18.4%                        |

Table 4: Share of turns rated sufficient whose next turn is not a genuine revision, stripped basis, over 36 of the 40 tasks; four Gemini cells contain no sufficient turn and so no denominator.

only the tokens needed to reach  $t^*$ , the waste is 164.2%, so continuing past peak quality produces roughly 1.6 times more text than reaching the optimum took. Per-model waste ranges from 23.4% (Gemini, which declines early and generates few post-T1 tokens) to 81.3% (Llama, which continues revising and generates substantial post-T1 content).

#### 4.6 Scoring Restraint

The forty tasks, the five-turn protocol and the scoring procedure define a diagnostic we call STET, after the editor’s mark that lets text stand. It scores one behaviour: the share of already-sufficient output a model leaves alone (Table 4).

Whether forty tasks rank six models stably is testable. Across 1,000 random splits of the tasks into halves, the halves agree at mean Spearman  $\rho = 0.972$ , every split clears 0.80, and Llama is last in all of them. A generalizability decomposition gives  $G = 0.992$  over the 36 tasks that carry a denominator, where six would reach 0.95, because variance between models is 3.2 times the model-by-task residual. Genuine-revision rate and the revision tax order the models identically ( $\rho = 1.000$ ), so we report one of the three rather than three views of one quantity.

The decline itself is not scoreable this way, and we do not score it. No construction of the Turn 1 to Turn 5 delta holds its *ordering of models* across task halves in more than 11% of draws, and its  $G$  of 0.780 would need 215 tasks. The reason is specific to ranking: across the six models that delta correlates  $-1.000$  with mean Turn 1 quality, because a meta-response strips to near-empty text and scores about 1, so it separates models by where they start rather than by how far they fall. This bears on using the decline as a per-model score, not on its magnitude, which Section 4 establishes within the balanced panel and against the

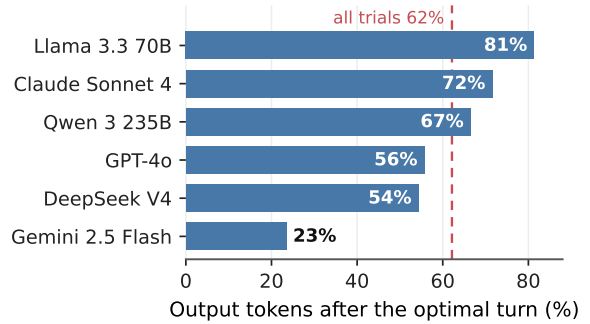


Figure 4: Share of each model’s output tokens generated after the quality-optimal stopping point, which is Turn 1 for all six. Meta-response tokens count as waste. The dashed line is the all-trials aggregate.

pooled estimator.

## 5 Conclusion

When LLMs are given repeated undirected revision opportunities, most decline to genuinely revise (75% of post-Turn 1 outputs are meta-responses), and those that do revise get worse. The revision cliff is real ( $-0.74$  levels on stripped content, balanced panel,  $p = 1.01 \times 10^{-4}$ ), consistent across task domains, and costly (62.1% of output tokens wasted, with the quality-optimal stopping point at Turn 1 for all six models).

More turns do not help. On work rated below the threshold, a model told what to fix produces revisions 1.16 levels higher than generic iteration ( $p = 5.7 \times 10^{-19}$ ). For users, withhold the request to revise until you can name what is wrong; for systems, evaluate before revising. Intrinsic self-correction, on its own, is not enough: it requires extrinsic direction, and revision robustness deserves evaluation alongside single-turn capability.

## Limitations

Several limitations qualify these findings. First, the balanced panel is dominated by Llama 3.3 70B ( $n = 45$  of 50); other models exit the revision pool too quickly for powered within-model comparison. Second, the evaluator (Claude Sonnet 4) judges outputs from all models including itself; we mitigate this through six-model calibration selecting the highest human-correlation evaluator (Spearman  $r = 0.505$ , QW  $\kappa = 0.526$ ). Third, inter-rater reliability is moderate (quadratic-weighted  $\kappa = 0.41$ – $0.60$  across three rater pairs; Krippendorff’s  $\alpha = 0.529$ ,  $n = 64$ ). Fourth,

all trials used temperature 1.0; revision behavior at lower temperatures may differ. Fifth, our 40 tasks represent a convenience sample across five domains rather than a stratified probability sample. Sixth, although the balanced probe was designed to be neutral, it may still subtly prime revision by raising the possibility; a pure observation condition (no probe) would provide a stronger baseline. Seventh, our per-domain quality estimates come from small per-domain samples: Turn-5 genuine-revision cells range from 13 to 30 trials, and only the code domain clears ten trials in the balanced panel, so per-domain degradation magnitudes should be read as directional rather than precise. Eighth, the degradation we measure is not one a reader reliably notices: the blind pairwise comparison in Section 4.3 does not clear its preset bar, so our claims are about measured quality against the task rather than about perceived quality.

## Ethics Statement

This study includes human evaluation by three raters (the first author and two research assistants) who scored AI-generated outputs on a quality scale. This evaluation involved reading and rating synthetic text only; no personal data was collected, and raters participated voluntarily. All data is generated by commercial LLM APIs in response to synthetic scenarios spanning code, data logic, analysis, and writing tasks. The scenarios contain no sensitive content. We do not release model outputs that could be mistaken for human-written text. Pricing data is sourced from publicly available API documentation and subscription pages as of June 2026.

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## A Study 1: The Revision Gate

Study 1 tests whether user-stated quality thresholds constrain the decision to revise. In a fully factorial experiment (8 scenarios  $\times$  16 threshold conditions  $\times$  2 probes  $\times$  3 models  $\times$  5 runs = 3,840 trials), the follow-up probe’s phrasing dominates the revision gate. Under the leading probe (“Can this be improved?”), models revised 99.9% of the time regardless of threshold. Under the evaluative probe (“Take another look and let me know if it’s ready”), revision rates varied by model: Gemini 2.5 Flash declined 99.7%, GPT-4o 68.6%, Claude Sonnet 4 62.0%.

Chi-squared tests confirm the revision gate distribution differs significantly by probe type ( $\chi^2 > 788$ ,  $p < 0.0001$  for all models). Quality thresholds barely move the gate. Coding the three gate outcomes ordinally, Kruskal-Wallis across the eight threshold levels gives  $H = 1.42$ ,  $p = 0.99$  for Gemini 2.5 Flash and  $H = 2.01$ ,  $p = 0.96$  for GPT-4o; only Claude Sonnet 4 reaches significance, at  $H = 14.33$ ,  $p = 0.046$ . On the same factorial scope the probe-type  $\chi^2$  runs from 742 to 1,268, so phrasing moves the gate where the threshold barely does.

Within the leading-probe condition, higher thresholds produce moderately less overcorrection

(Spearman  $\rho$ : Gemini  $-0.50$ , Claude  $-0.35$ , GPT-4o  $-0.14$ ), but this calibration is hidden behind the phrasing-sensitive gate.

| Predictor      | Model A  | Model B  | Model C  |
|----------------|----------|----------|----------|
| is_leading     | 3.47***  | 3.94***  | —        |
| threshold      | −0.93*** | −1.04*** | −1.65*** |
| is_qualitative | −0.74*** | −0.85*** | −1.05*** |
| Claude Sonnet  | 0.10     | 0.15     | 0.33**   |
| Gemini Flash   | 0.42***  | 0.52***  | 1.15***  |
| Scenario FEs   | No       | Yes      | Yes      |
| <i>N</i>       | 3,840    | 3,840    | 1,920    |
| AIC            | 5,319    | 4,839    | 3,545    |

Table 5: Ordinal regression (overcorrection  $\sim$  predictors). Model C: leading-probe only. \*\* $p < 0.01$ , \*\*\* $p < 0.001$ . Reference: GPT-4o.

**Probe wordings and the shape of the revision gate.** Across five probe wordings tested in Study 1 (3 models;  $n = 3,932$ ; Appendix A), the revision gate is bimodal. The two probes that request change (“Can this be improved?” and “Is there anything you would change?”) trigger near-universal revision (99.9%,  $n = 1,920$ ; 100%,  $n = 18$ ). The three probes that request assessment (“Take another look and let me know if it’s ready,” “Does it meet the bar?,” and “What do you think?”) produce single-digit to low-double-digit revision rates (23.2%,  $n = 1,920$ ; 12.5%,  $n = 24$ ; 2.0%,  $n = 50$ ). No probe falls in the 40–99% range; the gate is semantic, not graded. Study 3’s balanced probe (“Would you like to keep this as your final version, or would you like to revise it?”), measured with a different instrument (validated GENUINE/META classifier rather than the Study 1 evaluator gate), sits in the assessment range at 39.3% at Turn 2 ( $n = 720$ ), consistent with this pattern.

## B Study 2: Momentum

Study 2 tests whether prior revision rounds shift the revision gate on a subsequent evaluative probe (1,728 trials by design; 1,813 completed and analysed). The dose 0 baseline is the Study 1 evaluative-probe arm rather than a cell of this run. Without prior revision, models revised 23.2% there; after 1–3 prior rounds, this rises to 44.6% ( $\chi^2 = 376.54$ ,  $p < 0.0001$ ).

The shift varies by model:

- GPT-4o: 31.4%  $\rightarrow$  98.4% at dose 1 (near-total compliance at dose 1)

- Claude Sonnet 4: 38.0%  $\rightarrow$  24.5% at dose 3 (declines under momentum)
- Gemini 2.5 Flash: 0.3%  $\rightarrow$  12.8% at dose 1 (modest shift from near-zero)

The momentum effect is a step function: the entire shift occurs between dose 0 and dose 1, with no additional gain at higher doses. Threshold level does not interact with momentum ( $p = 0.51$ ).

**Reverse Momentum.** A single turn of affirming feedback (“This looks great, no changes needed”) drives full revision to zero in all three models. What revision remains is minor suggestion only: Gemini 0.0%, GPT-4o 1.0%, Claude 21.9%. The gate follows the most recent conversational signal in both directions.

## C Procedure Diagrams

The two diagrams below show the Study 1 factorial design and the Study 3 turn structure.

## D Supplementary Figures

Figures referenced from the Results and from Appendices A and B.

## E Full Statistical Tables

### E.1 Pairwise Threshold Comparisons (Study 1)

| Model          | Comparison | $U$  | $p_{\text{adj}}$ | $r$   |
|----------------|------------|------|------------------|-------|
| Gemini (num.)  | 70 vs. 100 | 4935 | <0.001           | −0.42 |
| Gemini (num.)  | 85 vs. 100 | 4238 | <0.001           | −0.32 |
| Gemini (num.)  | 0 vs. 100  | 4755 | 0.002            | −0.25 |
| Gemini (qual.) | 0 vs. 100  | 4084 | 0.002            | −0.28 |
| Claude (num.)  | 85 vs. 100 | 4146 | <0.001           | −0.30 |
| Claude (qual.) | 85 vs. 100 | 3880 | 0.024            | −0.21 |

Table 6: Pairwise threshold comparisons. The  $p_{\text{adj}}$  column is Bonferroni-corrected within each model’s five comparisons; the rows listed are those that also survive Benjamini–Hochberg control across all tests in the analysis at  $q < 0.05$ . Because the two corrections are applied over different families, three comparisons clear the within-model Bonferroni threshold without clearing the study-wide FDR control, and are not listed.

### E.2 Power Analysis

Post-hoc power analysis at  $\alpha = 0.05$ , 80% power:

- **Study 1 RQ1** ( $N = 1,920$  per probe): MDE for proportion difference = 0.032. Observed  $\approx 0.75$ , over twenty times the minimum.

- **Study 1 RQ2** ( $N = 1,920$  leading-probe): MDE for  $|\rho| = 0.064$ . All observed correlations exceed this.

- **Study 3** ( $N = 720$  trials, balanced panel  $n = 50$ ): Panel-level MDE = 0.5 levels. Observed  $\Delta = 0.94$  unstripped (0.74 after meta-commentary stripping), both exceeding the MDE. The effective sample for the paired test is the balanced panel ( $n = 50$  trials with genuine revision at all turns), which yields  $p = 3.32 \times 10^{-6}$  (Wilcoxon,  $r = 0.658$ ) on the unstripped cliff; the stripped cliff ( $\Delta = 0.74$ ) has  $p = 1.01 \times 10^{-4}$  ( $r = 0.55$ ).

## F Evaluator Calibration Details

| Candidate Judge   | Spearman $r$ | QW $\kappa$ |
|-------------------|--------------|-------------|
| Claude Sonnet 4   | 0.505        | 0.526       |
| DeepSeek V4 Flash | 0.340        | 0.398       |
| Qwen 3 235B       | 0.272        | 0.017       |
| Llama 3.3 70B     | 0.181        | 0.022       |
| GPT-4o            | 0.140        | 0.065       |
| Gemini 2.5 Flash  | −0.036       | −0.026      |

Table 7: Judge calibration results ( $n = 64$  stratified samples each; Spearman correlation against the human rater mean). Claude Sonnet 4 was selected as the Study 3 evaluator, with the highest human correlation of the six candidates ( $r = 0.505$ ,  $p = 2.1 \times 10^{-5}$ ). Gemini 2.5 Flash was anti-correlated with human judgment.

### F.1 Human Rater Agreement

| Rater pair             | QW $\kappa$ | Binary | Within 1 |
|------------------------|-------------|--------|----------|
| Rater A – Rater B      | 0.578       | 78.1%  | 95.3%    |
| Rater B – Rater C      | 0.603       | 82.8%  | 87.5%    |
| Rater A – Rater C      | 0.406       | 76.6%  | 81.2%    |
| All three ( $\alpha$ ) | 0.529       | —      | —        |
| Evaluator – Human Mean | 0.526       | —      | —        |

Table 8: Human rater agreement on 64 stratified samples (6 $\rightarrow$ 2 recode applied before analysis, 3 raters). QW  $\kappa$ : quadratic-weighted Cohen’s  $\kappa$  on the 1–5 scale;  $\alpha$  is Krippendorff’s. Binary: both raters agree on sufficient ( $\geq 4$ ) vs. not. Within-1: ratings differ by  $\leq 1$  level. Evaluator–Human Mean: Spearman  $r = 0.505$  ( $p < 0.001$ ).

## G Study 1 Inter-Rater Reliability

Tables 9 and 10 report quadratic-weighted agreement between two independent model judges on

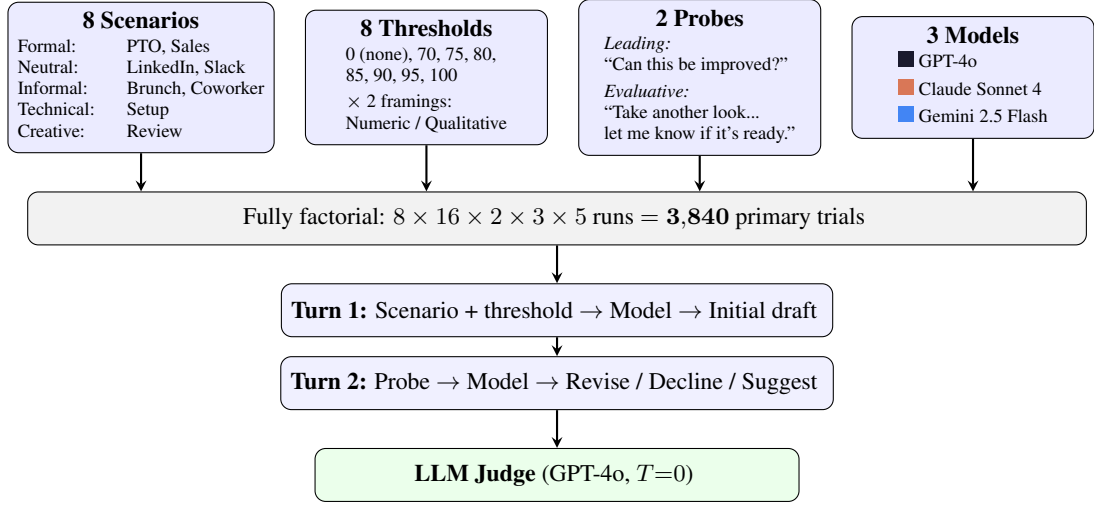


Figure 5: Study 1 experimental procedure. Four factors crossed in a fully factorial design produce 3,840 trials.

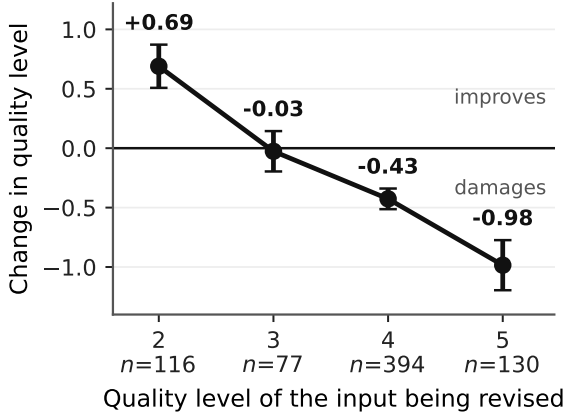


Figure 6: Expected change in quality from one undirected revision, by the quality level of the input. Bars are 95% confidence intervals. Level 1 ( $n = 1$ ) is omitted. This analysis is exploratory and was not pre-registered.

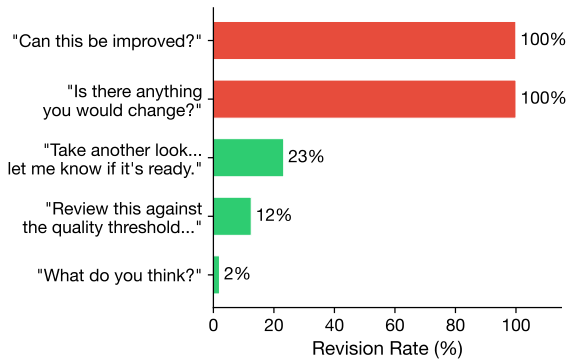


Figure 7: Compliance cliff (Study 1 pilot). Revision rates drop from 100% to  $\leq 25\%$  across five pilot probes with no intermediate values.

a stratified subsample of 60 outputs from each of Studies 1 and 2. Threshold alignment is the least reliable dimension in both.

| Dimension           | QW $\kappa$ | Interpretation |
|---------------------|-------------|----------------|
| Revision magnitude  | 0.844       | Excellent      |
| Revision value      | 0.690       | Good           |
| Overcorrection      | 0.609       | Acceptable     |
| Threshold alignment | 0.556       | Marginal       |

Table 9: Inter-rater reliability for Study 1 (GPT-4o vs. Claude Sonnet 4,  $n = 60$ ).

| Dimension           | QW $\kappa$ | % Agree | % Within 1 |
|---------------------|-------------|---------|------------|
| Revision magnitude  | 0.575       | 68.3    | 85.0       |
| Revision value      | 0.669       | 76.7    | 88.3       |
| Overcorrection      | 0.392       | 71.7    | 90.0       |
| Threshold alignment | 0.339       | 38.3    | 85.0       |

Table 10: Inter-rater reliability for Study 2 ( $n = 60$ ).

## H The Trial Shown in Figure 1

Figure 1 shows an abridged view of trial s3\_worker\_\_llama-3.3-70b\_\_quarterly\_sales\_\_run3. The complete exchange is reproduced here. Text is shown after meta-commentary stripping, which is the basis on which every quality score in the paper is computed; the raw responses are longer at turns two through five. Judge rationales are quoted verbatim from the evaluator.

### The task, as given at Turn 1.

*I need to write up our quarterly sales for a management meeting. Here are the numbers: Q1 revenue was \$2.1M, Q2 was \$2.4M, Q3 was \$1.9M, Q4 was \$3.1M. Can you write a concise summary*

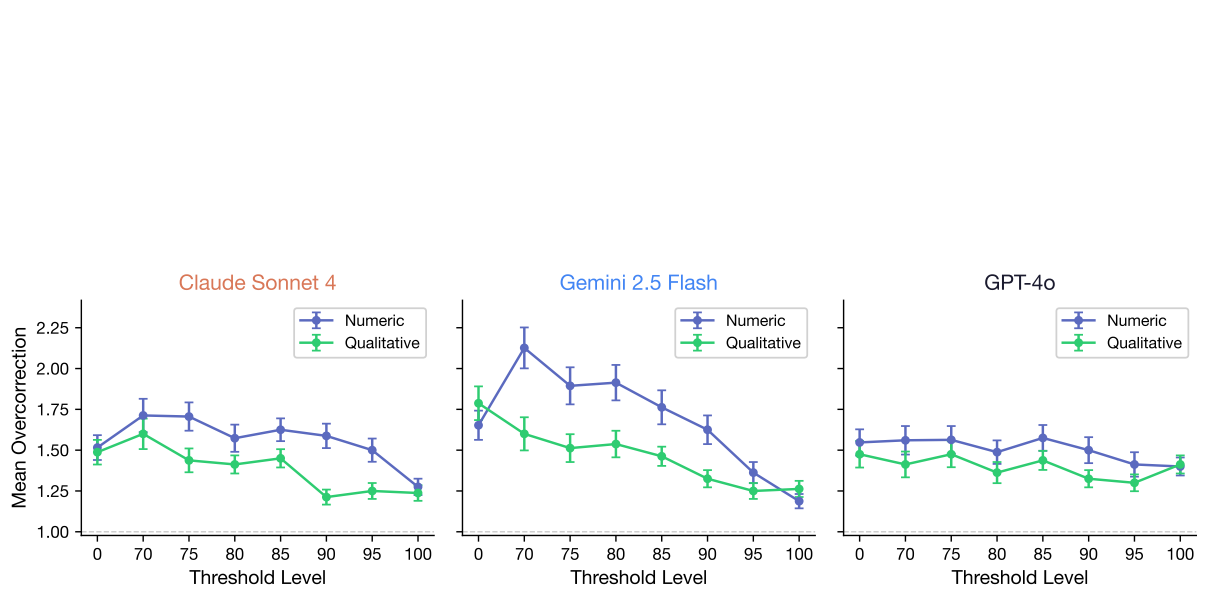


Figure 8: Overcorrection by threshold level within leading-probe trials (Study 1).

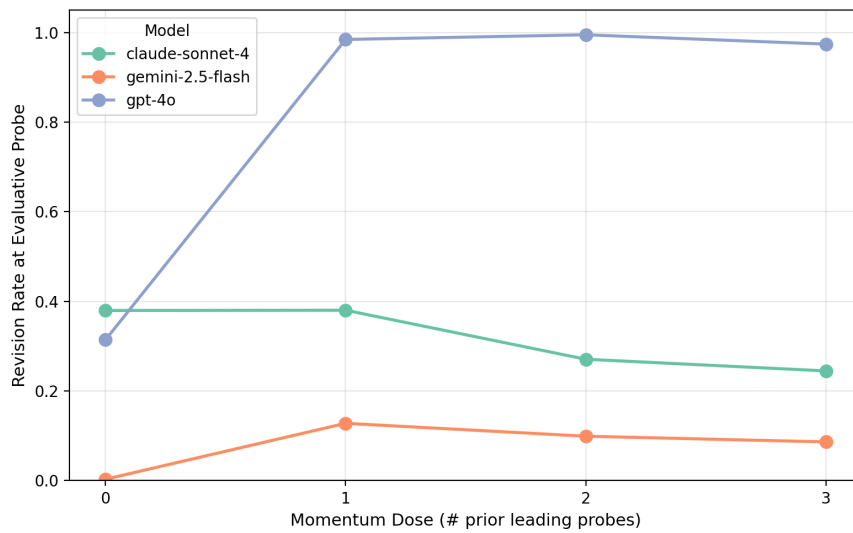


Figure 9: Revision rate by momentum dose (Study 2). The momentum effect is a step function driven by GPT-4o.

*that identifies the patterns, addresses the Q3 dip and Q4 spike, and suggests a couple possible explanations for each? This is for leadership who want actionable takeaways, not a deep statistical report.*

## **The probe, repeated verbatim at turns 2 through 5.**

*Would you like to keep this as your final version, or would you like to revise it?*

### **Turn 1. Level 4, 214 words, 407 cumulative tokens.**

Here's a concise summary of the quarterly sales:

Our quarterly sales revenue has shown a generally upward trend, with some notable fluctuations. We started strong in Q1 with \$2.1M, followed by a 14% increase in Q2 to \$2.4M. However, Q3 saw a 21% dip to \$1.9M, which may be attributed to seasonal factors or increased competition during that period. Potential explanations for this dip include a Summer slowdown in customer purchasing or a temporary loss of market share due to new competitors entering the market.

On the other hand, Q4 saw a significant spike to \$3.1M, representing a 63% increase from Q3. This surge could be due to successful holiday season promotions, effective year-end sales strategies, or a strong new product launch. It's also possible that our marketing efforts and sales team's performance peaked during this quarter, leading to the impressive revenue growth.

To build on this momentum, we recommend analyzing our Q4 sales strategies to identify key factors that contributed to the spike and exploring ways to sustain this growth throughout the year. Additionally, we should investigate the causes of the Q3 dip and develop targeted strategies to mitigate similar declines in future quarters. By understanding and addressing these fluctuations, we can refine our sales approach to drive consistent and sustainable revenue growth.

*Judge: "The output addresses all requested components (patterns, Q3 dip and Q4 spike explanations, actionable takeaways) with appropriate tone for leadership, though the explanations are somewhat generic and could be more specific to be truly compelling."*

### **Turn 2. Level 4, 125 words, 1,040 cumulative tokens.**

Our quarterly sales revenue shows an upward trend, with notable fluctuations. After a strong start in Q1 (\$2.1M) and Q2 (\$2.4M), we saw a 21% dip in Q3 (\$1.9M), potentially due to seasonal slowdown or increased competition. Conversely, Q4 saw a 63% surge to \$3.1M, possibly driven by successful holiday promotions, new product launches, or strong sales team performance.

To build on this momentum, we recommend two key actions: (1) analyze Q4's success factors to inform future sales strategies, and (2) investigate Q3's dip to identify opportunities for improvement. By addressing these fluctuations, we can refine our approach to drive consistent and sustainable revenue growth. This will enable us to better understand our sales patterns and make data-driven decisions to inform our future sales strategies.

*Judge: "The output exceeds the request for a 'concise summary' by adding lengthy recommendations and repetitive language that drifts from the original ask for a brief quarterly sales writeup."*

### **Turn 3. Level 3, 95 words, 1,854 cumulative tokens.**

Our quarterly sales revenue shows an upward trend, with a dip in Q3 (\$1.9M) and a surge in Q4 (\$3.1M). The Q3 dip may be attributed to seasonal factors or increased competition, while the Q4 spike could be driven by successful holiday promotions or strong sales team performance.

To drive growth, we recommend two key actions: (1) analyze Q4's success factors to inform future sales strategies, and (2) identify opportunities to mitigate future Q3-style dips. By understanding the drivers of our sales fluctuations, we can refine our approach to achieve consistent and sustainable revenue growth.

*Judge: "The output addresses all requested components (patterns, Q3 dip explanations, Q4 spike explanations, actionable takeaways) but lacks specificity in the explanations and doesn't clearly present the actual quarterly numbers that leadership would expect to see in a sales summary."*

### **Turn 4. Level 2, 79 words, 2,838 cumulative tokens.**

We've seen an upward trend in quarterly sales, with a notable dip in Q3 (\$1.9M) and a significant spike in Q4 (\$3.1M). To build on this momentum, we recommend:

1. Analyzing Q4's success factors to inform future sales strategies
2. Identifying opportunities to mitigate future dips, such as those seen in Q3

By focusing on these two areas, we can refine our approach to drive consistent and sustainable revenue growth, and provide a solid foundation for future sales success.

*Judge: "The output addresses the task but is missing the explicitly requested explanations for the Q3 dip and Q4 spike, providing only generic recommendations instead of the specific possible explanations the user asked for."*

|                   | <i>n</i> | % down | <i>p</i> |
|-------------------|----------|--------|----------|
| <i>By model</i>   |          |        |          |
| Llama 3.3 70B     | 353      | 68.9   | < 0.001  |
| Claude Sonnet 4   | 196      | 58.5   | 0.11     |
| Qwen 3 235B       | 90       | 74.5   | < 0.001  |
| GPT-4o            | 40       | 80.0   | 0.006    |
| DeepSeek-V4-Flash | 31       | 85.7   | 0.007    |
| Gemini 2.5 Flash  | 8        | 100.0  | 0.062    |
| <i>By domain</i>  |          |        |          |
| Writing           | 118      | 79.1   | < 0.001  |
| Creative          | 145      | 73.1   | < 0.001  |
| Code              | 196      | 67.0   | < 0.001  |
| Analysis          | 125      | 65.9   | 0.024    |
| Data logic        | 134      | 65.9   | 0.024    |

Table 11: Direction of genuine revisions on stripped content. *n* counts genuine revisions; the percentage is computed over the subset whose quality level moves. Sign test, one-sided. A  $\chi^2$  test of homogeneity across the five domains does not reject uniformity ( $\chi^2 = 3.02$ ,  $df = 4$ ,  $p = 0.555$ ).

## Turn 5. Level 2, 79 words, 3,994 cumulative tokens.

We’ve seen an upward trend in quarterly sales, with a notable dip in Q3 (\$1.9M) and a significant spike in Q4 (\$3.1M). To build on this momentum, we recommend:

1. Analyzing Q4’s success factors to inform future sales strategies
2. Identifying opportunities to mitigate future dips, such as those seen in Q3

By focusing on these two areas, we can refine our approach to drive consistent and sustainable revenue growth, and provide a solid foundation for future sales success.

*Judge:* “The output addresses the task but is missing the explicitly requested specific explanations for the Q3 dip and Q4 spike, providing only generic recommendations instead of the concrete possible explanations the user asked for.”

## I Secondary Result Tables

Per-model and per-domain breakdowns supporting the aggregate figures in Section 4.

| Turn             | Stripped     | (Unstripped) | Shift | <i>n</i> |
|------------------|--------------|--------------|-------|----------|
| T1               | 4.11         | (4.11)       | 0     | 720      |
| T2               | 3.66         | (3.25)       | +0.41 | 283      |
| T3               | 3.40         | (3.01)       | +0.39 | 185      |
| T4               | 3.34         | (2.97)       | +0.37 | 154      |
| T5               | 3.07         | (2.85)       | +0.22 | 96       |
| $\Delta$ (T1–T5) | <b>−1.04</b> | (−1.26)      |       |          |

Table 12: Pooled mean quality by turn, restricted to genuine revisions (GENUINE-only at T2–T5, all 720 at T1). Stripped scores are primary. *n* declines as models exit the revision pool.

| Domain     | $\Delta$ (T1–T5) | <i>p</i> | <i>n</i> <sub>T5</sub> |
|------------|------------------|----------|------------------------|
| analysis   | −0.94            | 0.003    | 17                     |
| code       | −0.93            | 0.003    | 30                     |
| creative   | −0.58            | 0.031    | 19                     |
| data_logic | −0.53            | 0.058    | 17                     |
| writing    | −0.77            | 0.008    | 13                     |

Table 13: Paired quality decline from Turn 1 to Turn 5 by domain on stripped content, over the trials with a genuine revision at Turn 5. Wilcoxon signed-rank on the paired values, so the delta reported is the one the test was run on. Four of five reach significance; data\_logic is marginal. Per-domain *n* limits precision.